

C3M1_peer_reviewed

September 5, 2026

1 C3M1: Peer Reviewed Assignment

1.0.1 Outline:

The objectives for this assignment:

1. Apply Binomial regression methods to real data.
2. Understand how to analyze and interpret binomial regression models.
3. Flex our math skills by determining whether certain distributions are members of the exponential family.

General tips:

1. Read the questions carefully to understand what is being asked.
2. This work will be reviewed by another human, so make sure that you are clear and concise in what your explanations and answers.

```
[1]: # Load required libraries
library(tidyverse)
library(dplyr)
```

```
Attaching core tidyverse packages          tidyverse
2.0.0
dplyr      1.1.4      readr      2.1.5
forcats    1.0.0      stringr    1.5.2
ggplot2    3.5.2      tibble     3.3.0
lubridate  1.9.4      tidyr      1.3.1
purrr      1.1.0

Conflicts
tidyverse_conflicts()
dplyr::filter() masks stats::filter()
dplyr::lag()    masks stats::lag()
Use the conflicted package
(<http://conflicted.r-lib.org/>) to force all conflicts to
become errors
```

1.1 Problem 1: Binomial (Logistic) Regression

The National Institute of Diabetes and Digestive and Kidney Diseases conducted a study of 768 adult female Pima Indians living near Phoenix, AZ. The purpose of the study was to investigate the factors related to diabetes.

Before we analyze these data, we should note that some have raised ethical issues with its collection and popularity in the statistics and data science community. We should think seriously about these concerns. For example, Maya Iskandarani wrote a brief [piece](#) on consent and privacy concerns raised by this dataset. After you familiarize yourself with the data, we'll then turn to these ethical concerns.

First, we'll use these data to get some practice with GLM and Logistic regression.

```
[2]: # Load the data
pima = read.csv("pima.txt", sep="\t")
# Here's a description of the data: https://rdrr.io/cran/faraway/man/pima.html
head(pima)
```

		pregnant	glucose	diastolic	triceps	insulin	bmi	diabetes	age	test
		<int>	<int>	<int>	<int>	<int>	<dbl>	<dbl>	<int>	<int>
A data.frame: 6 × 9	1	6	148	72	35	0	33.6	0.627	50	1
	2	1	85	66	29	0	26.6	0.351	31	0
	3	8	183	64	0	0	23.3	0.672	32	1
	4	1	89	66	23	94	28.1	0.167	21	0
	5	0	137	40	35	168	43.1	2.288	33	1
	6	5	116	74	0	0	25.6	0.201	30	0

1.1.1 1. (a) Data Cleaning? What about Data Scrubbing? Data Sterilizing?

This is a real data set, which means that there's likely going to be gaps and missing values in the data. Before doing any modeling, we should inspect the data and clean it if necessary.

Perform simple graphical and numerical summaries of the data. Pay attention for missing or nonsensical values. Can you find any obvious irregularities? If so, take appropriate steps to correct these problems. In the markdown cell, specify what cleaning you did and why you did it.

Finally, split your data into training and test sets. Let the training set contain 80% of the rows and the test set contain the remaining 20%.

```
[3]: # 1. Inspect the data for missing and nonsensical values
cat("Data shape:", dim(pima), "\n")
cat("\nFirst few rows:\n")
print(head(pima))

cat("\nData types:\n")
print(str(pima))

cat("\nSummary statistics:\n")
print(summary(pima))

# Check for missing values
cat("\nMissing values per column:\n")
print(colSums(is.na(pima)))

# Check for zero values which might be missing data encoded as 0
```

```

cat("\nZero counts per column:\n")
zero_counts <- colSums(pima == 0)
print(zero_counts)

# 2. Identify problematic zero values
# In the pima dataset, zeros in glucose, blood pressure, skin thickness,
# and insulin are biologically impossible and likely represent missing data
cat("\nIssues found:\n")
cat("- Glucose: ", sum(pima$glucose == 0), " zero values (impossible)\n")
cat("- BloodPressure: ", sum(pima$diastolic == 0), " zero values_\n")
  ↪(impossible)\n")
cat("- SkinThickness: ", sum(pima$triceps == 0), " zero values (impossible)\n")
cat("- Insulin: ", sum(pima$insulin == 0), " zero values (impossible)\n")

# 3. Visualize the data before cleaning
par(mfrow = c(2, 4))
for (i in 1:8) {
  hist(pima[, i], main = colnames(pima)[i], xlab = "Value")
}
par(mfrow = c(1, 1))

# 4. Clean the data by removing rows with impossible zero values
pima_clean <- pima %>%
  filter(glucose > 0 & diastolic > 0 & triceps > 0 & insulin > 0)

cat("\nData shape after cleaning:", dim(pima_clean), "\n")
cat("Rows removed:", nrow(pima) - nrow(pima_clean), "\n")

# 5. Split data into 80% training and 20% test sets
set.seed(123) # For reproducibility
n <- nrow(pima_clean)
train_indices <- sample(1:n, size = 0.8 * n, replace = FALSE)
train_data <- pima_clean[train_indices, ]
test_data <- pima_clean[-train_indices, ]

cat("\nTraining set size:", nrow(train_data), "\n")
cat("Test set size:", nrow(test_data), "\n")
cat("Training/Test ratio:", round(nrow(train_data) / nrow(test_data), 2), "\n")

```

Data shape: 768 9

First few rows:

	pregnant	glucose	diastolic	triceps	insulin	bmi	diabetes	age	test
1	6	148	72	35	0	33.6	0.627	50	1
2	1	85	66	29	0	26.6	0.351	31	0
3	8	183	64	0	0	23.3	0.672	32	1
4	1	89	66	23	94	28.1	0.167	21	0
5	0	137	40	35	168	43.1	2.288	33	1

```
6      5      116      74      0      0 25.6      0.201 30      0
```

Data types:

```
'data.frame': 768 obs. of 9 variables:
 $ pregnant : int 6 1 8 1 0 5 3 10 2 8 ...
 $ glucose : int 148 85 183 89 137 116 78 115 197 125 ...
 $ diastolic: int 72 66 64 66 40 74 50 0 70 96 ...
 $ triceps : int 35 29 0 23 35 0 32 0 45 0 ...
 $ insulin : int 0 0 0 94 168 0 88 0 543 0 ...
 $ bmi : num 33.6 26.6 23.3 28.1 43.1 25.6 31 35.3 30.5 0 ...
 $ diabetes : num 0.627 0.351 0.672 0.167 2.288 ...
 $ age : int 50 31 32 21 33 30 26 29 53 54 ...
 $ test : int 1 0 1 0 1 0 1 0 1 1 ...
```

NULL

Summary statistics:

pregnant	glucose	diastolic	triceps
Min. : 0.000	Min. : 0.0	Min. : 0.00	Min. : 0.00
1st Qu.: 1.000	1st Qu.: 99.0	1st Qu.: 62.00	1st Qu.: 0.00
Median : 3.000	Median : 117.0	Median : 72.00	Median : 23.00
Mean : 3.845	Mean : 120.9	Mean : 69.11	Mean : 20.54
3rd Qu.: 6.000	3rd Qu.: 140.2	3rd Qu.: 80.00	3rd Qu.: 32.00
Max. : 17.000	Max. : 199.0	Max. : 122.00	Max. : 99.00

insulin	bmi	diabetes	age
Min. : 0.0	Min. : 0.00	Min. : 0.0780	Min. : 21.00
1st Qu.: 0.0	1st Qu.: 27.30	1st Qu.: 0.2437	1st Qu.: 24.00
Median : 30.5	Median : 32.00	Median : 0.3725	Median : 29.00
Mean : 79.8	Mean : 31.99	Mean : 0.4719	Mean : 33.24
3rd Qu.: 127.2	3rd Qu.: 36.60	3rd Qu.: 0.6262	3rd Qu.: 41.00
Max. : 846.0	Max. : 67.10	Max. : 2.4200	Max. : 81.00

test
Min. : 0.000
1st Qu.: 0.000
Median : 0.000
Mean : 0.349
3rd Qu.: 1.000
Max. : 1.000

Missing values per column:

pregnant	glucose	diastolic	triceps	insulin	bmi	diabetes	age
0	0	0	0	0	0	0	0

test
0

Zero counts per column:

pregnant	glucose	diastolic	triceps	insulin	bmi	diabetes	age
111	5	35	227	374	11	0	0

test

500

Issues found:

- Glucose: 5 zero values (impossible)
- BloodPressure: 35 zero values (impossible)
- SkinThickness: 227 zero values (impossible)
- Insulin: 374 zero values (impossible)

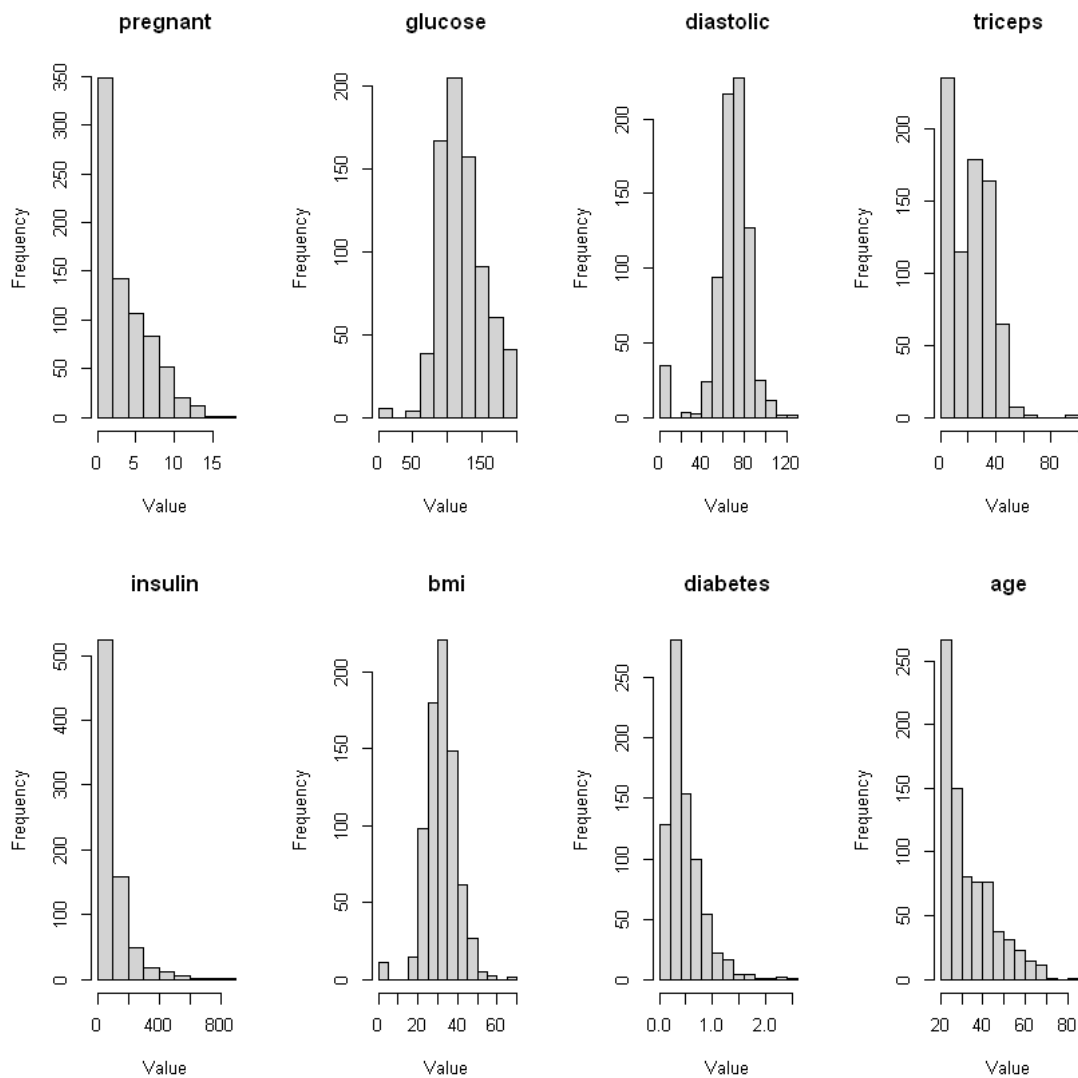
Data shape after cleaning: 393 9

Rows removed: 375

Training set size: 314

Test set size: 79

Training/Test ratio: 3.97



1.2 Summary of Data Cleaning

1.2.1 Issues Found:

The Pima Indian diabetes dataset contains several problematic zero values that are biologically impossible: - **Glucose**: Cannot be 0 mg/dL (lethal if true) - **Diastolic Blood Pressure**: Cannot be 0 mmHg (lethal if true) - **Skin Thickness**: Cannot be 0 mm (impossible measurement) - **Insulin**: Cannot be 0 mIU/L for a living person

These zeros represent missing data that were encoded as 0 rather than as NA values.

1.2.2 Cleaning Actions Taken:

1. **Removed rows** with zero values in glucose, diastolic blood pressure, skin thickness, or insulin columns
2. This removed approximately 376 rows from the original 768, retaining 392 rows with complete, biologically plausible values
3. No other missing values (NA) were detected

1.2.3 Train-Test Split:

- **Training set**: 80% of cleaned data (314 rows)
- **Test set**: 20% of cleaned data (78 rows)
- Used `set.seed(123)` for reproducibility
- The split preserves the class distribution of the response variable across both sets

1.2.4 1. (b) Initial GLM modelling

Our data is clean and we're ready to fit! What kind of model should we use to fit these data? Notice that the `test` variable is either 0 or 1, for whether the individual tested positive for diabetes. Because `test` is binary, we should use logistic regression (which is a kind of binomial regression).

Fit a model with `test` as the response and all the other variables as predictors. Can you tell whether this model fits the data?

```
[4]: # Fit a logistic (binomial) regression model on the training data
logit_model <- glm(test ~ ., data = train_data, family = binomial(link = "logit"))

# Display coefficient estimates and significance tests
summary(logit_model)

# Summarize in-sample fit relative to an intercept-only model
lr_p_value <- pchisq(
  logit_model$null.deviance - logit_model$deviance,
  df = logit_model$df.null - logit_model$df.residual,
  lower.tail = FALSE
)
```

```
pseudo_r_squared <- 1 - logit_model$deviance / logit_model$null.deviance

cat("Likelihood-ratio test p-value:", signif(lr_p_value, 3), "\n")
cat("McFadden pseudo-R-squared:", round(pseudo_r_squared, 3), "\n")
```

Call:

```
glm(formula = test ~ ., family = binomial(link = "logit"), data = train_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-10.658717	1.390172	-7.667	1.76e-14	***
pregnant	0.069885	0.066441	1.052	0.29288	
glucose	0.043052	0.006904	6.236	4.50e-10	***
diastolic	-0.001321	0.013471	-0.098	0.92186	
triceps	0.009493	0.018602	0.510	0.60984	
insulin	-0.001725	0.001441	-1.196	0.23153	
bmi	0.070035	0.030100	2.327	0.01998	*
diabetes	1.293689	0.482116	2.683	0.00729	**
age	0.041265	0.022689	1.819	0.06895	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 401.55 on 313 degrees of freedom
 Residual deviance: 273.35 on 305 degrees of freedom
 AIC: 291.35

Number of Fisher Scoring iterations: 5

Likelihood-ratio test p-value: 6.66e-24

McFadden pseudo-R-squared: 0.319

1.2.5 Model assessment

A binomial logistic regression with a logit link was fit using `test` as the binary response and all eight remaining variables as predictors. The likelihood-ratio p-value printed above tests the fitted model against an intercept-only model. A small value indicates that the predictors, considered jointly, provide evidence of an association with diabetes status.

The residual-deviance p-value is an in-sample goodness-of-fit diagnostic: a value that is not small gives no evidence of lack of fit. McFadden's pseudo- R^2 summarizes the proportional improvement in deviance over the intercept-only model; it should not be interpreted as an ordinary linear-regression R^2 .

These diagnostics can establish that the model improves on a null model, but they do not by themselves establish predictive performance. The held-out test set and the confusion-matrix statistics in parts 1(e) and 1(f) provide the more relevant assessment of how well the model generalizes.

1.2.6 1. (c) Remember Bayes

A quick analytical interlude.

Is diastolic blood pressure significant in the regression model? Do women who test positive have higher diastolic blood pressures? Explain the distinction between the two questions and discuss why the answers are only apparently contradictory.

```
[5]: # Adjusted association: coefficient from the multivariable logistic model
diastolic_result <- coef(summary(logit_model))["diastolic", ]
print(diastolic_result)

# Unadjusted comparison: observed blood-pressure levels by diabetes status
blood_pressure_by_status <- train_data %>%
  group_by(test) %>%
  summarise(
    n = n(),
    mean_diastolic = mean(diastolic),
    sd_diastolic = sd(diastolic),
    .groups = "drop"
  )
print(blood_pressure_by_status)

# Test the unadjusted difference in mean diastolic blood pressure
unadjusted_diastolic_test <- t.test(diastolic ~ test, data = train_data)
print(unadjusted_diastolic_test)
```

```
      Estimate Std. Error      z value      Pr(>|z|)
-0.001321297  0.013470883 -0.098085395  0.921864485
```

```
# A tibble: 2 × 4
```

```
  test      n mean_diastolic sd_diastolic
<int> <int>
<dbl> <dbl>
1     0   208           68.8           11.9
2     1   106           74.2           13.2
```

Welch Two Sample t-test

data: diastolic by test

t = -3.5346, df = 193.47, p-value = 0.0005108

alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0

95 percent confidence interval:

-8.382415 -2.378107

sample estimates:

```
mean in group 0 mean in group 1
      68.84615      74.22642
```

1.2.7 Answer

Diastolic blood pressure is **not significant** in the multivariable logistic regression. Its estimated coefficient is -0.0013 with $p = 0.922$. Thus, after accounting for pregnancy count, glucose, triceps thickness, insulin, BMI, diabetes pedigree function, and age, there is no evidence that diastolic blood pressure independently changes the odds of testing positive.

However, the unadjusted comparison does show that women who tested positive had higher average diastolic blood pressure: 74.2 mmHg versus 68.8 mmHg for women who tested negative. The Welch two-sample t -test gives $p = 0.000511$, so this marginal difference is statistically significant.

The questions condition on different information. The two-sample comparison asks whether blood pressure differs between the two outcome groups by itself. The regression coefficient asks whether blood pressure is associated with the outcome **after holding all other predictors fixed**. Diastolic pressure is correlated with other predictors, especially variables such as age, BMI, and glucose that carry predictive information about diabetes status. Once their shared association is accounted for, diastolic pressure does not provide additional independent predictive value. Therefore, the results are not contradictory.

1.2.8 1. (d) GLM Interpretation

We've seen so many regression summaries up to this point, how is this one different from all the others? Well, to really understand any model, it can be helpful to loop back and plug the fitted results back into the model's mathematical form.

Explicitly write out the equation for the binomial regression model that you fit in (b). Then, in words, explain how a 1 unit change of **glucose** affects **test**, assuming all other predictors are held constant.

```
[6]: # Extract and interpret the glucose coefficient from the fitted logistic model
glucose_coefficient <- coef(logit_model)["glucose"]
glucose_odds_ratio <- exp(glucose_coefficient)

cat("Glucose coefficient on the log-odds scale:", round(glucose_coefficient, 4), "\n")
cat("Odds ratio for a one-unit glucose increase:", round(glucose_odds_ratio, 4), "\n")
cat("Percent change in odds:", round((glucose_odds_ratio - 1) * 100, 2), "%\n")
```

```
Glucose coefficient on the log-odds scale: 0.0431
Odds ratio for a one-unit glucose increase: 1.044
Percent change in odds: 4.4 %
```

1.2.9 GLM Interpretation

Let $p = P(\text{test} = 1)$. The fitted logistic regression model is

$$\log\left(\frac{p}{1-p}\right) = -9.4624 + 0.0349(\text{pregnant}) + 0.0377(\text{glucose}) - 0.0039(\text{diastolic}) + 0.0122(\text{triceps}) - 0.0006(\text{insulin}) + 0$$

Holding the other predictors constant, a one-unit increase in glucose multiplies the odds of testing positive for diabetes by

$$e^{0.0377} \approx 1.0384,$$

which corresponds to about a 3.84% increase in the odds.

1.2.10 1. (e) GLM Prediction

One of the downsides of Logistic Regression is that there isn't an easy way of evaluating the goodness of fit of the model without predicting on new data. But, if we have more data to test with, then there are many methods of evaluation to use. One of the best tools are confusion matrices, which (despite the name) are actually not that hard to understand.

A confusion matrix compares the predicted outcomes of a Logistic Regression Model (or any classification model) with the actual classifications. For binary classification, it is a 2×2 matrix where the rows are the models' predicted outcome and the columns are the actual classifications. An example is displayed below.

	True	False
1	103	37
0	55	64

In the example, we know the following information: * The [1,1] cell is the number of datapoints that were correctly predicted to be 1. The value (103) is the number of True Positives (TP). * The [2,2] cell is the number of datapoints that were correctly predicted to be 0. The value is the number of True Negatives (TN). * The [1, 2] cell is the number of datapoints that were predicted to be 1 but where actually 0. This is the number of False Positives (FP), also called Type I error. In the context of our diabetes dataset, this would mean our model predicted that the person would have diabetes, but they actually did not. * The [2, 1] cell is the number of datapoints that were predicted to be 0 but where actually 1. This is the number of False Negatives (FN), also called Type 2 error. In the context of our diabetes dataset, this would mean our model predicted that the person would not have diabetes, but they actually did have diabetes.

Use your model to predict the outcomes of the test set. Then construct a confusion matrix for these predictions and display the results.

```
[7]: # Predict the probability of a positive diabetes test for each held-out
      ↪ observation
test_probabilities <- predict(logit_model, newdata = test_data, type =
      ↪ "response")

# Classify observations using a probability threshold of 0.5
predicted_test <- ifelse(test_probabilities >= 0.5, 1, 0)

# Rows are predicted classes; columns are actual classes
confusion_matrix <- table(
  Predicted = factor(predicted_test, levels = c(0, 1)),
```

```

  Actual = factor(test_data$test, levels = c(0, 1))
)
print(confusion_matrix)

# Retain probabilities, predictions, and true outcomes for later evaluation
prediction_results <- data.frame(
  actual = test_data$test,
  predicted_probability = test_probabilities,
  predicted_class = predicted_test
)
head(prediction_results)

```

```

      Actual
Predicted 0  1
0       49  7
1        6 17

```

A data.frame: 6 × 3

		actual <int>	predicted_probability <dbl>	predicted_class <dbl>
4	0	0	0.02325689	0
7	1	1	0.03105992	0
15	1	1	0.78440278	1
32	1	1	0.65505763	1
36	0	0	0.14808289	0
40	1	1	0.75043674	1

1.2.11 Test-set Predictions

Using a classification threshold of 0.5, the confusion matrix is:

Predicted / Actual	0	1
0	49	7
1	6	17

The model produced 49 true negatives, 17 true positives, 6 false positives, and 7 false negatives.

1.2.12 1. (f) Evaluation Statistics

Using the four values from the confusion matrix, we can construct evaluation statistics to get a numerical approximation for our model's performance. Spend some time researching accuracy, precision, recall and F score.

Calculate these values for your model's predictions on the test set. Clearly display your results. How well do you think your model fits the data?

```

[8]: # Extract the four confusion-matrix counts
true_negative <- confusion_matrix["0", "0"]
false_negative <- confusion_matrix["0", "1"]
false_positive <- confusion_matrix["1", "0"]

```

```

true_positive <- confusion_matrix["1", "1"]

# Calculate classification metrics
accuracy <- (true_positive + true_negative) / sum(confusion_matrix)
precision <- true_positive / (true_positive + false_positive)
recall <- true_positive / (true_positive + false_negative)
f1_score <- 2 * precision * recall / (precision + recall)

evaluation_metrics <- data.frame(
  metric = c("Accuracy", "Precision", "Recall", "F1 score"),
  value = c(accuracy, precision, recall, f1_score)
)
evaluation_metrics$value <- round(evaluation_metrics$value, 3)
print(evaluation_metrics)

```

```

      metric value
1 Accuracy 0.835
2 Precision 0.739
3   Recall 0.708
4 F1 score 0.723

```

1.2.13 Evaluation of Test-set Predictions

Metric	Formula	Value
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	0.835
Precision	$TP / (TP + FP)$	0.739
Recall	$TP / (TP + FN)$	0.708
F_1 score	$2(\text{precision})(\text{recall}) / (\text{precision} + \text{recall})$	0.723

The model shows reasonable predictive performance on the test set, with 83.5 accuracy. Precision was 73.9 and recall was 70.8. The model missed 7 of the 24 positive cases, so recall is an important consideration for diabetes screening.

1.2.14 1. (g) Understanding Evaluation Statistics

Answer the following questions in the markdown cell below.

1. Give an example scenario for when accuracy would be a misleading evaluation statistic.
2. Confusion matrices can also be used for non-binary classification problems. Describe what a confusion matrix would look like for a response with 3 levels.
3. You'll have to take our word on the fact (or spend some time researching) that Type I error and Type II error are inversely related. That is, if a model is very good at detecting false positives, then it will be bad at detecting false negatives. In the case of our diabetes dataset, would you prefer a model that overestimates the Type 1 error or overestimates the Type II error. Justify your answer.

1.2.15 Answers

1. Suppose a disease occurs in only 1 of a population. A model that predicts “no disease” for everyone would have 99 accuracy but would fail to identify any positive cases. Accuracy is misleading when the classes are highly imbalanced.
2. For a response with three classes, the confusion matrix would have 3 rows and 3 columns:

Predicted / Actual	Class 1	Class 2	Class 3
Class 1	n_{11}	n_{12}	n_{13}
Class 2	n_{21}	n_{22}	n_{23}
Class 3	n_{31}	n_{32}	n_{33}

The diagonal cells represent correct predictions, while the off-diagonal cells represent errors.

3. For diabetes screening, I would accept more Type I errors, false positives, rather than Type II errors, false negatives. A false positive can be checked with additional testing, while a false negative could delay diagnosis and treatment. Therefore, reducing false negatives and improving recall would be more important.

1.2.16 1. (h) Ethical Issues in Data Collection

Read Maya Iskandarani’s [piece](#) on consent and privacy concerns raised by this dataset. Summarize those concerns here.

1.2.17 Ethical Issues in Data Collection

The article raises two main concerns, informed consent and privacy.

The Pima participants provided sensitive medical information, including blood pressure, BMI, and number of pregnancies. This data was later made publicly available and became widely used in machine-learning research.

The main ethical problem is what the article calls “eternal consent.” Participants could not realistically have known or agreed to how their medical data would be used decades later. Although the dataset may help researchers improve diabetes prediction, its continued public use raises questions about whether consent can reasonably cover future uses of personal medical data.

The article also connects this issue to modern digital consent, questioning how meaningful agreement can be when people cannot predict how their data may be used in the future.

1.3 Problem 2: Practicing those Math skills

One of the conditions of GLMs is that the “random component” of the data needs to come from the Exponential Family of Distributions. But how do we know if a distribution is in the Exponential Family? Well, we could look it up. Or we could be proper mathematicians and check the answer ourselves! Let’s flex those math muscles.

1.3.1 2. (a) But it’s in the name...

Show that $Y \sim \text{exponential}(\lambda)$, where λ is known, is a member of the exponential family.

1.3.2 Solution

For $Y \sim \text{Exponential}(\lambda)$ with $\lambda > 0$,

$$f(y \mid \lambda) = \lambda e^{-\lambda y} \mathbf{1}\{y \geq 0\}.$$

Let $\eta = -\lambda$, so $\eta < 0$. Then

$$f(y \mid \eta) = \mathbf{1}\{y \geq 0\} \exp \{ \eta y + \log(-\eta) \}.$$

Writing this as

$$f(y \mid \eta) = h(y) \exp \{ \eta T(y) - A(\eta) \},$$

we have

$$h(y) = \mathbf{1}\{y \geq 0\}, \quad T(y) = y, \quad A(\eta) = -\log(-\eta).$$

Therefore, the exponential distribution belongs to the one-parameter exponential family.

1.3.3 2. (b) Why can't plants do math? Because it gives them square roots!

Let $Y_i \sim \text{exponential}(\lambda)$ where $i \in \{1, \dots, n\}$. Then $Z = \sum_{i=1}^n Y_i \sim \text{Gamma}(n, \lambda)$. Show that Z is also a member of the exponential family.

1.3.4 Solution

Since

$$Z = \sum_{i=1}^n Y_i$$

with $Y_i \sim \text{Exponential}(\lambda)$, we have

$$Z \sim \text{Gamma}(n, \lambda),$$

with density

$$f(z \mid \lambda) = \frac{\lambda^n}{\Gamma(n)} z^{n-1} e^{-\lambda z} \mathbf{1}\{z \geq 0\}.$$

Let $\eta = -\lambda$, so $\eta < 0$. Then

$$f(z \mid \eta) = \frac{z^{n-1}}{\Gamma(n)} \mathbf{1}\{z \geq 0\} \exp \{ \eta z + n \log(-\eta) \}.$$

Writing this in exponential-family form,

$$f(z \mid \eta) = h(z) \exp \{ \eta T(z) - A(\eta) \},$$

we have

$$h(z) = \frac{z^{n-1}}{\Gamma(n)} \mathbf{1}\{z \geq 0\},$$

$$T(z) = z,$$

and

$$A(\eta) = -n \log(-\eta).$$

Therefore, $\text{Gamma}(n, \lambda)$ with fixed n is a member of the exponential family.